AI agent Discovery

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AI Discovery Current State

- "Do something new"
  - Add auth, security, etc.
- "Handle it via the network"
  - DNS, SIP-like registration with carrier networks, etc
- "Handle it via applications"
  - Model cards & well-known URL

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AI Discovery Current State

- "Do something new" -> What happens if the IETF takes too long?
  - Add auth, security, etc.
- "Handle it via the network" -> What happens if the metadata is too rich?
  - DNS, SIP-like registration with carrier networks, etc
- "Handle it via applications" -> What if application owners create dysfunction?
  - Model cards & well-known URL

DNS-AID

"Handle it via the network"

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                    Agent Capabilities

┌──────────────────────────┐     ┌──────────────────────────┐
│      DISCOVERY           │     │     APPLICATION          │
│                          │     │                          │
│  - SVCB (INDEX)          │     │  - TRAVEL                │
│      IP                  │     │  - COST                  │
│      PROTOCOL            │     │  - ATTESTATIONS          │
│      PORT                │     │     TRAINING DATA        │
│                          │     │     CERTIFICATIONS       │
│  -WELL KNOWN AGENTS      │     │     ACCEPTABLE USE       │
│     SVCB RRs             │     │  - TO BE SPECIFIED...    │
│                          │     │                          │
└──────────────────────────┘     └──────────────────────────┘

       This draft                       Other drafts
```

AI Discovery Status

- Service (chat) and domain (example.com) combine into a well-known entry point: chat._agents.example.com
- SVCB Params to express application data (eg cost, modality, well-known URI / model card location)

- Coauthor group believes *most often* agents know BOTH (status = 1). Many times they know OR (status = 2 | 3), few times they know NOR (status = 0)
  - Therefore, the majority use case of discovery is solvable via DNS

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AI Agent Client
wants to discover:   foobar._mcp._agents.example.com
(mcp = service, foobar = agent/capability, example.com = trusted domain)

  Agent Client                Resolver / Cache         Root/TLD/Auth DNS
        |                           |                          |
        |   1. SVCB Query           |                          |
        |   (Recursive)             |                          |
        |---------------------------|                          |
        |                           |                          |
        |                           | 2. Query Root            |
        |                           |--------------------------|
        |                           |                          |
        |                           | 3. Referral to .com NS   |
        |                           |<-------------------------|
        |                           |                          |
        |                           | 4. Query .com TLD        |
        |                           |--------------------------|
        |                           |                          |
        |                           | 5. Referral to auth NS   |
        |                           |<-------------------------|
```

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|                              |                              |
|                              | 6. Query Auth NS for         |
|                              |    foobar._mcp._agents       |
|                              |------------------------------>|
|                              |                              |
|                              | 7. SVCB RRSet Response       |
|                              |    (DNSSEC-signed)           |
|                              |<------------------------------|
|                              |                              |
|   8. SVCB Response           |                              |
|   (cached + validated)       |                              |
|<------------------------------|                              |
|                              |                              |
|   Response includes:         |                              |
|   • Service target           |                              |
|   • ALPN protocol(s)         |                              |
|   • Port number              |                              |
|   • IPv4/IPv6 hints          |                              |
|   • DNSSEC signatures        |                              |
|   • Capabilities (cost,      |                              |
|      modalities, etc)        |                              |
```


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DNS AID

- Facilitating agent to agent discovery in a decentralized manner that won't lead to internet fragmentation, nor sovereignty issues with equal opportunity for ownership between circuit/network and app/developer personas.

- Democratizing the use and access of AI solutions, disadvantaging walled garden approaches that lock consumers into proprietary registries. Delivered as a known entry point to an organization (e.g. index._agents.example.com), also opportunities for individual agent records.
  - The contents / transactions between an index and agents, standardization efforts of model cards, etc. are reserved for AI working group activity, but this discovery method is an integration source for *any* protocol development (TBD)

- Bonus: using existing technologies for rapid adoption.
  - Create known subdomains (_agents.example.com) where SVCB records advertise agents (chat._mcp._example.com) and their capabilities through parameters (e.g. JSON model card's URI, cost, location, etc).
  - Capability for index servers of all agents at example.com.
  - Capability for registry services of all agents or indexes of agents at example.com and example2.com.

- Cons: potentially imperfect discovery (some use cases excluded).

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Technical summary overview

- Versions:
  - V0 – all DNS capabilities (DANE etc)
  - V1 – feedback is to incorporate model cards, do less with DNS (stripped out many capability recommendations)
- Guidance provided to operators to prevent DNS abuse (spam, record duplication, zones etc.)
- Metadata (metrics required in discovery)
  - JSON model card URI, cost, location, modalities, etc. (yet to be standardized)
- IANA to reserve a leaf attribute zone (e.g. _agents.example.com), SVCB params (e.g. cost, location, attestations, JSON model card URI, etc.) which need to be standardized by IETF WGs over time.
  - Strip out the 'mandatory' discovery fields to accelerate discovery, decentralize, facilitate registry-to-registry communications, etc.

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Example

- Tencent, Alibaba Cloud, Xiaohongshu, Xianyu deploy OpenClaw overnight
- OpenClaw uses hardcoded core gateway, channels, and tools in TypeScript user
  - agents are configured per-workspace via local prompts/config,
  - capabilities extended via a plugin/skills system plus the ClawHub registry
- Discovery today is intra-ecosystem: What an OpenClaw agent “is” and “can do” is learned from local config + skills/registry conventions, not from a standardized, interoperable discovery protocol or DNS namespace.
- DNS-AID standardizes where and how to publish agent endpoints and capabilities in DNS (SVCB/HTTPS/TXT/TLSA + DNSSEC/DCV), so any framework can discover and verify agents across domains in a uniform way.

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IETF work?

- See DNSOP
  - If DNS is to be used at all within AI agent discovery (SHOULD), this work needs thorough IETF review & collaboration, contributions, etc.
  - An extension / special use of DNS-SD with _agents leaf zone (RFC 7558)
  - Utilize SVCB records + params to express metadata (RFC 9460/1)
  - WEBBOTAUTH, DMSC, ANS, DA-ITN, etc.
- New WG most likely (preferred by coauthors)
  - INTAREA
    - Agent2Agent (pending charter + inclusion of discovery?)
    - AIPROTO / discovery standalone?

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Work to date and future plans

- AI Discovery Side Meeting (IETF 124 Montreal)
  - https://github.com/nicknacnic/BANDAID/blob/main/Side%20Meeting%20for%20AI%20Discovery.pdf
- Many in the scope of discovery, this author group the DNS AID draft is most relevant, various problem space drafts
- Mostly discussing within Agent2Agent mailing list, open source communities (ie Balázs' agentcommunity: https://github.com/agentcommunity/agent-interface-discovery), and DNS providers
- 125 Plans: *HACKATHON*, CATALIST BOF, side meetings, etc.
- 126: BOF specifically for discovery (in / out of AIPROTO/A2A WG etc.)
- Other: Liaison statements / collaboration with ITU-T SG17

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